

Health Screening AI Project

Weekly Progress Report · Week 1

One page · Friday by 6 PM · Specific numbers and links, not summaries

Intern name: Saugata Malakar	Role: Healthcare Privacy · LLM · Multimodal AI
Week / date: Week 1 · 22 May 2026	GitHub username: saugata-malakar
W&B / Drive link: PII_Field_Map_DPDP_Saugata.xlsx — submitted to Drive (link below in Section B)	

✓ A. Work completed this week (4–5 bullets)

Deliverables: PII field map + DPDP gap analysis + anonymisation architecture design. All three Week 1 outputs complete.

- **PII field map completed across all 26 database tables — 152 columns classified.** Three-tier classification: **47 Direct Identifiers (DI)** (patient_id, full_name, Aadhaar, phone, photo_url, consultation free_text_notes, reports); **30 Quasi-Identifiers (QI)** (date_of_birth, village_name, district, ip_address, weight_kg, height_cm, diabetes_duration_yrs); **75 Non-Sensitive (NS)** (ICD codes, Wagner grade, model_confidence, audit timestamps, clinical boolean flags). Each column mapped with anonymisation action and governing DPDP Act section. Submitted to Drive.
- **DPDP Act 2023 compliance gap analysis completed — 10 gaps identified (G-01 to G-10).** Three non-negotiables from the brief formally documented: **G-02 (\$5 — data localisation):** Cloud Run region unconfirmed, must be asia-south1. **G-03 (\$6(6) — 72-hour erasure):** deletion cascade across all 26 tables not yet built. **G-04 (\$8(4) — sensitive data):** wound_images stored as plaintext paths; aadhaar_number column unencrypted. All 10 gaps assigned to owner and target week.
- **Anonymisation architecture designed on paper — pending PI review before Week 2 coding.** Design covers: HMAC-SHA256 + rotating 32-byte salt for all patient/user/worker ID pseudonymisation; age generalised to 5-year bands (0–4, 5–9, ...); village_name stripped from all exports, district retained; k-anonymity ≥ 5 verification against QI combination (age band + sex + district); FK-respecting 26-table deletion cascade order documented (sessions → audit_logs → notifications → prescriptions → ... → patients). Draft submitted — awaiting PI sign-off.
- **Critical §16 cross-border data transfer risk identified and escalated (Gap G-08).** Gemini 1.5 Pro Vision API defaults to us-central1 (USA). Week 4 multimodal prompts will transmit wound photographs to US Google servers — a potential §16 DPDP violation without Data Protection Board approval. No confirmed asia-south1 Vertex AI Gemini endpoint found in public documentation. Raised as PI-level decision; on-prem inference fallback to be evaluated.
- **Resource repository compiled — 50+ references indexed.** Covers: official DPDP Act 2023 + Rules 2025 texts (MeitY), 6 DPDP academic papers, 6 anonymisation theory papers (k-anon, differential privacy, pseudonymisation), 8 FL/Flower references, 7 clinical NLP/spaCy tutorials, 5 RAG/LangChain/FAISS docs, 5 OWASP security guides. Delivered as searchable HTML hub with filter-by-category.

📁 B. Code and files submitted (exact links, not 'see GitHub')

Week 1 deliverables are documentation artefacts submitted to Drive. GitHub code submissions begin Week 2.

- **Drive — PII_Field_Map_DPDP_Saugata.xlsx:** Sheet 1: 26-table PII field map (152 columns, DI/QI/NS classification, anonymisation action, DPDP section reference). Sheet 2: DPDP Act 2023 gap analysis (G-01 to G-10, current status, action needed, owner, week). **Link:** [Drive link — to be pasted on submission]
- **Drive — resource_guide.html:** Searchable resource hub, 50+ curated references. Filterable by: Law/Policy, Research Papers, Tools/Libraries, Videos/Courses, Blogs/Guides, GitHub/Code, Docs. **Link:** [Drive link — to be pasted on submission]
- **GitHub note — code repositories begin Week 2:** feat/anonymisation (anonymisation module, HMAC-SHA256, k≥5 verification), feat/erasure (26-table deletion cascade), feat/export (data export endpoint with anonymisation layer). Repository access to be confirmed by PI on Day 1 of Week 2.

⚠ C. Problems faced (specific — what broke and what you tried)

- **Real database schema not yet received — Week 2 is blocked without it.** PII field map was built against a 26-table schema inferred from the technical brief and platform stack (PostgreSQL + Cloud Run + React Native). Actual column names, data types, and FK relationships may differ from assumptions. Attempted to derive schema from the brief's stack hints and team deliverables (e.g., Sharif's Wagner Grade columns, Kousttav's leprosy model outputs). However, the anonymisation module CANNOT be written against assumed column names — Week 2 start is blocked until real migration files are received.
- **Gemini API endpoint region unconfirmed — §16 DPDP blocker for Week 4.** Reviewed Google Cloud documentation: Gemini 1.5 Pro in Vertex AI does not list an asia-south1 (Mumbai) endpoint as generally available. Tried to find an India-specific Gemini inference endpoint in Cloud console documentation — none confirmed. Current default routes to us-central1, which would constitute a cross-border personal data transfer under §16 of the DPDP Act.
- **Anonymisation architecture requires PI sign-off before any code is written.** Brief explicitly states: 'Get it reviewed by the PI before Week 2.' Architecture document is complete and submitted this week. No code written. If PI feedback arrives after Monday, Week 2 coding start will be delayed — the brief's required sequence cannot be shortcut.

🔔 D. Help needed from PI (who, what, by when)

- **PI — actual database migration files or pg_dump --schema-only — needed by Monday.** The anonymisation module (HMAC-SHA256, k-anon verification) and erasure pipeline (26-table cascade) cannot be written against assumed column names. Alembic migration folder or a single pg_dump --schema-only output is sufficient. **Hard blocker for Week 2 start.**
- **PI — Cloud Run region and Gemini API routing confirmation — needed by Wednesday.** Gap G-02: confirm Cloud Run deployment and GCS bucket region = asia-south1 (§5 data localisation). Gap G-08: confirm whether Gemini 1.5 Pro Vision can be routed through an Indian/APAC endpoint, or whether an on-prem inference fallback (e.g., self-hosted LLaVA or a Vertex AI endpoint in asia-south1) is required for Week 4 multimodal integration. §16 violation risk if unresolved.
- **PI — review and written sign-off on anonymisation architecture — needed by Monday EOD.** Brief requirement: 'Design the anonymisation architecture on paper before writing code. Get it reviewed by the PI before Week 2.' Draft submitted this week. A short written 'approved' or a list of required changes by Monday EOD is sufficient to unblock Week 2 coding.

- **Sahil (Backend) — GitHub repository access and dev DB credentials — needed by Monday.** Brief states access granted Day 1. Needed to run the erasure pipeline end-to-end on dev DB and verify zero records remain — the Week 2 test evidence requires live DB access.

 E. Targets for next week (3–4 measurable goals with numbers)

- **Anonymisation module — unit-tested, $k \geq 5$ verified — submitted to GitHub by Friday.** HMAC-SHA256 pseudonymisation for patient_id, doctor_id, worker_id, contact_id with rotating 32-byte salt. Age generalised to 5-year bands. Village_name stripped. k-anonymity ≥ 5 verified on QI combination (age_band + sex + district). ≥ 10 unit tests covering re-identification attempts, salt rotation, and edge cases. GitHub branch: feat/anonymisation.
- **Erasure pipeline — all 26 tables, tested and verified on dev DB — submitted to GitHub by Friday.** Python script covering all 26 tables in correct FK dependency order. Test procedure: create synthetic patient + linked records → run erasure → query all 26 tables → assert zero rows remain. Deletion log submitted as test evidence. GitHub branch: feat/erasure.
- **Data export endpoint integrated with anonymisation layer — verified by Friday.** POST /export/patient passes all query results through the anonymisation module before returning. Verified: no raw patient_id, name, Aadhaar, phone, or village_name appears in any export response. GitHub branch: feat/export.
- **PII field map corrections — updated with real schema and re-submitted to Drive by Monday EOD (contingent on schema delivery).** If schema received Monday morning, all 26 table entries updated with real column names and data types by Monday end of day.

Self-assessment	PI notes (returned by Monday)
 On track All three Week 1 deliverables completed: 26-table PII field map (152 columns), DPDP gap analysis (10 gaps, G-01 to G-10), and anonymisation architecture design. Two critical dependency blockers identified and escalated to PI: real DB schema (Week 2 blocker) and Gemini API region (Week 4 \$16 risk). No code written this week — per brief, architecture review must precede implementation.	